

1 Query, Contract & Plan Space



User



Query Q

What are the termination rights and notice periods under the supply contract?
...

Evidence Contract (Budget & Guarantees)



Risk $\leq \alpha$



Cost $\downarrow$



Citation (optional)



Freshness (optional)

Plan Space $\mathcal{P}$ (RAG Pipeline)



Retrieve



Rerank



Fuse



Generate

Model Families

Qwen

DeepSeek

GLM

Gemma

Llama

...

Heterogeneous RAG Plans $\mathcal{P} = \{p_1, \dots, p_P\} \cup \{\perp\}$

p_1

p_2

...

p_P

$\perp$

Decline



Decline action $\perp$ has loss 0 and cost $c_{\perp}$.

2 Ledger Gate (Validity)



Ledger Balance

$$B_{t-1} = \alpha(t-1) - V_{t-1}$$

α : risk budget per round

V_{t-1} : cumulative audit loss



IF $B_{t-1} \geq 1 - \alpha$



Gate OPEN



Proceed to
Action LP

ELSE



Gate CLOSED



Decline $\perp$
with $\ell_t = 0$



Pathwise guarantee

$V_t \leq \alpha t$ for every t

✓ No δ (no probability level)

✓ No exchangeability

✓ Independent of $|\mathcal{P}|$

3 Cost Optimisation (Learning)

1



Action LP: $\min \sum_a w_a c_a$
s.t. $\sum_a w_a \tilde{r}_a \leq \alpha$ over $\mathcal{P} \cup \{\perp\}$

2



Optimistic risk: $\tilde{r}_a = \hat{r}_a + \frac{\kappa}{\sqrt{n_a}}$

3



Sample action $a \sim w$

4



Execute plan or decline

5



Audit loss $\ell_t \in [0, 1]$

6



Update ledger
 $B_t \leftarrow B_{t-1} + \alpha - \ell_t$
and learn estimates



Regret $O\left(\sqrt{\frac{P \log T}{T}}\right)$ vs offline OPT



(1) Validity by accounting

- Pathwise, no statistics



(2) Cost by learning

- Statistics only for money



(3) Decline as first-class action

- Every $\alpha > 0$ feasible